

A145 – Gravitational Constant G from ξ

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.1 The Central Formula

[K] In this theory the gravitational constant is not an independent quantity but follows from ξ and a mass:

$$G = \frac{\xi^2}{4 m_{\text{char}}} \quad (\text{natural units}).$$

This is a *derivation*, not a bridge. G is not absorbed as a unit factor (that would be A080), but expressed as a function of the single parameter and a characteristic mass. The difference from A130 is decisive: there α was in the end calibrated; here the *structure* of G is determined from ξ .

[B] The equivalent form

$$\xi = 2\sqrt{G m_{\text{char}}}$$

shows the statement from the other side: ξ , the tetrahedral packing quantity from A020, is up to the factor the square root of the product of gravitational strength and characteristic mass. Gravity and geometry are thereby not two things but two readings of ξ .

This is a statement about structure only in natural units (A085). In SI it would be an equation between incommensurable units.

[K] For context: in natural units $G = 1$, exactly as $c = \hbar = k_B = \alpha = 1$ (A085, A130). The SI value $6.674 \cdot 10^{-11}$ is bookkeeping. The formula $G = \xi^2/(4m)$ therefore does not say what native value G has – that is one – but how the *SI dress* of G composes from ξ and a mass. This is the same bookkeeping as for α , only for a dimensioned rather than a dimensionless quantity.

.2 Why This Is a Particularly Direct Derivation

[K] As with α (A130), there is no circularity at the core of the formula here either – for α the initial circularity is broken by the two-route over-determination of E_0 ; for G it is absent from

the outset: $G = \xi^2/(4m)$ does not use G anywhere to determine G . The only difference is directness – α needs the meeting of two routes, G does not because the formula simply does not contain G . A hidden calibration is not in the core here either, only in the SI dress (E_{char} , see below). The formula says: choose the characteristic mass – say m_e – and the gravitational strength follows from ξ .

[B] This is the actual resolution of the gravity problem in this theory, and it is stronger than what A140 states. A140 says: the QG problem does not arise because the geometry is pre-ordered. That is a *negative* argument. A145 says: G has a *positive* expression from ξ . The negative argument explains why quantisation is unnecessary; the positive explains where the numerical value comes from. This makes G the most direct ξ -manifestation in the corpus – not the most precisely tested (that is the lepton ladder A110 and the α over-determination A130), but the most immediate in form.

.3 The Dimension Bridge E_{char}

[K] The basic formula $G = \xi^2/(4m)$ is not yet dimensionally complete: in natural units G has dimension $[E^{-2}]$, but the right-hand side has $[E^{-1}]$. A factor $1/E_{\text{char}}$ is missing:

$$G_{\text{nat}} = \frac{\xi_0^2}{4m_e} \cdot \frac{1}{E_{\text{char}}}, \quad E_{\text{char}} \approx 28.4.$$

[K] The corpus provides a structural decomposition of E_{char} :

$$E_{\text{char}} = E_0 \cdot \left(\frac{4}{3}\right)^2 \cdot \frac{\pi}{\sqrt{2}} \cdot K_{\text{frak}} = 7.40 \cdot 1.778 \cdot 2.221 \cdot 0.986 = 28.8,$$

against the target value 28.4, i.e. +1.5 %.

The assessment – and it is reassuring. E_{char} carries two quantities that are not purely geometric elsewhere: E_0 (A130) and K_{frak} (A040). The factor $(4/3)^2$ is geometric, $\pi/\sqrt{2}$ likewise; E_0 and K_{frak} are not. But this concerns only the *SI numerical value* of G – and that is bookkeeping. How precisely E_{char} hits 28.4 is a statement about the scale conversion, not about the structure. As long as one works with ratios, the measured value of G is irrelevant; the structural relations are known and confirmed in SI and in the Standard Model. The *basic structure* $G = \xi^2/(4m)$ is unaffected by this and is the actual statement.

Cleanly separated: the statement “ G is proportional to ξ^2 and inversely proportional to a mass” is **[K]**. The precise numerical value via E_{char} is **[B]** with a calibrated component.

.4 The Same Diagnosis as for α , More Complex Dress

[K] G passes the same test by which A135 decides whether a quantity is fundamental: *can it be set to one by a consistent unit choice?* For G the answer is yes – in Planck units $G = c = \hbar = 1$. By the A135 diagnosis this means: G is not content of the theory but a conversion factor. It thereby stands on the same side as c , $\hbar$, k_B (physically one via duality) and α (one via normalisation) – all convention. The SI value $6.674 \cdot 10^{-11}$ is what this convention conceals, not a fundamental numerical value.

[K] And as with α , the content lies not in this value but in ξ . The bare coin is ξ ; G is – seen in SI light – the same coin, only dressed. That is the meaning of $G = \xi^2/(4m_{\text{char}})$: not a second degree of freedom but ξ in dimensioned form. Equivalent information content, not equal rank – exactly the resolution of the apparent tension from A135.

[K] The difference from α is only the *richness of the dress*, and it has a dimensional reason. In natural units α is dimensionless, $[E^0]$; its SI form therefore needs exactly one scale:

$$\alpha_{\text{SI}} = \xi \cdot E_0^2 \quad (\text{one scale, squared}).$$

G by contrast has dimension $[E^{-2}]$ in natural units (it is M_{Planck}^{-2}) and thereby ties mass to length to time. Its SI form can therefore not make do with one scale; it needs a mass *and* the intermediate scale E_{char} :

$$G_{\text{nat}} = \frac{\xi^2}{4 m_{\text{char}}} \cdot \frac{1}{E_{\text{char}}}.$$

E_{char} is itself multi-layered – $E_0 \cdot (4/3)^2 \cdot (\pi/\sqrt{2}) \cdot K_{\text{frak}}$ (see above) – whereas α makes do with the one E_0 . This is the “more complex SI dress”: not more content but more layers of the same content, forced by the richer dimension of G.

.5 What Counts: Exact Reconstructibility

[K] The central statement of this document is not how precisely the 0.5 % is hit, but that G in natural units can be set to one – and from this something exact follows. If G is a convention (one in Planck units), then its SI value is not an independent numerical value but *completely* fixed by the conversion:

$$G_{\text{SI}} = \frac{\hbar c}{M_{\text{Planck}}^2}.$$

Given ξ and one absolute scale, G_{SI} is thereby *exactly* determined – without a free parameter, without approximation. This is the same structure as $\alpha_{\text{SI}} = \xi E_0^2$: the dimensioned constant is the SI expression of ξ plus scale, and the reconstruction is exact, not approximate.

[K] The E_{char} decomposition is strictly separate from this. The exact E_{char} – the one that hits G_{SI} – is already fixed by ξ and the scale. What is 0.5 % off is not this exact quantity but the *attempt to express it by a closed formula* from known quantities ($E_0 \cdot (4/3)^2 \cdot (\pi/\sqrt{2}) \cdot K_{\text{frak}}$). The deviation measures the quality of this *analytical approximation of the dress*, not the reconstructibility itself. The reconstruction is exact; only its shortcut to a simple expression is not yet.

[STIPULATION] The priority is thereby correct: what distinguishes G from a genuinely fundamental constant is the settability to one and the resulting exact reconstructibility – and that is shown. The remaining E_{char} approximation is a question of closed form, not of structure, and carries no weight for the central statement.

.6 The Scale Hierarchy

[B] The formula orders three energy scales:

$$E_0 = 7.40 \text{ MeV} \ll E_{\text{char}} = 28.4 \ll E_{T0} = \frac{1}{\xi_0} = 7500.$$

E_0 is the electromagnetic scale (A130), $E_{T0} = 1/\xi$ the fundamental scale of the theory, E_{char} the intermediate scale at which gravity couples. That gravity sits at an *intermediate scale* is the structural statement behind its weakness: it couples far below the fundamental scale.

[S] Whether this hierarchy *quantitatively* explains the observed weakness of gravity – the forty orders of magnitude relative to electromagnetism – is not shown by this. It is a direction, not a calculation.

.7 Verification Script

- $G = \xi^2/(4m_e)$ and the reverse direction $\xi = 2\sqrt{Gm}$; the dimensional gap and its closure by E_{char} ; the structural decomposition of E_{char} with identification of the calibrated component; and the scale hierarchy: python/A_Serie_Skripte/a145_gravitationskonstante.py

.8 Symbol Glossary

Symbols used throughout the A-Series are explained in A010. Specific to this document:

Symbol	Meaning
m_{char}	Characteristic mass; limiting case of the self-consistency condition, $G = \xi^2/(4m_{\text{char}})$
G	Newtonian gravitational constant

.9 Open Questions

First, the characteristic mass is not uniquely fixed. The corpus uses m_e ; why the electron and not another mass is the characteristic one is not justified.

Second, E_{char} carries a calibrated component (see above). The basic structure $\xi^2/(4m)$ is unaffected by this; the precise numerical value is not. *This residual is explicitly not a measurement effect.* It is tempting to attribute the $\sim 0.5\%$ to the notorious measurement weakness of G – but the opposite is the case: G is measured to $2 \cdot 10^{-5}$ (about 0.002%), and even the large scatter between individual experiments only reaches $\sim 0.05\%$. The theory's residual is thus roughly 200 times larger than the measurement uncertainty and about 10 times larger than the experimental scatter; the inputs to the dress (E_0 via masses to $\sim 10^{-8}$, K_{frak} derived, the geometry factors exact) do not carry it either. The residual is thus structural – a residual of the E_{char} approximation – and must remain named as such; attributing it to measurement imprecision would be the same target-first assignment that A105 and A130 warn against.

Third, the residual is entangled with a Standard Model assumption – the mass definition. Measured lepton masses enter the dress ($E_0 = \sqrt{m_e m_\mu}$, $m_{\text{char}} = m_e$), and "mass" in the Standard Model is scheme-dependent: the measured values are *pole masses*, whereas a purely geometric construction could equally correspond to a running (MS-bar) mass. The difference between both schemes is of order $\alpha/\pi \approx 0.23\%$ per mass; since G depends on the masses via $G \sim 1/(m_e^{3/2} m_\mu^{1/2})$ with total mass power 2, this propagates to $\sim 0.46\%$ on G – *the same order as the residual*. This cuts both ways: it shows that the 0.5% *cannot* be cleanly read as a geometric defect because part of it may be a convention on the SM input side; but it does *not* show that the scheme effect closes the gap. That would only be shown by computing the pole-to-running shift for precisely these masses and this formula with sign and coefficient *forward* – an order-of-magnitude argument is not a derivation. Only this remains: the residual lies between geometric approximation and mass scheme, and both are of the same order.

Fourth, the gravitational dynamics are still missing. A145 gives the *value* of G from ξ ; it does not give the field equations from which the inverse-square law and motion would follow. This remains the large open point already named in A140.

.10 Supersedes

This document subsumes: Doc. 012 (gravitational constant, derivation part), Doc. 127 (gravity, Planck perspective). It supplements A140 with the positive derivation; A140 retains the bridge and QG argumentation. Incorporated: P40. New: the measurement of G by the one-test (Planck units $G = c = \hbar = 1$) and the dress comparison with α (dimensional reason for the richer SI dress).

.11 Use of AI Tools

This document was created with the assistance of AI tools. The questions, stipulations, and all substantive decisions come from the author; the tools formulate, compute, and create verification scripts. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.